

Predictive Parsing (2)

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This slide deck is still under active development. Expect changes before it is finalized.

Learning outcomes

- **After this lecture, you should be able to:**
- **derive parse trees from a BNF grammar**
- **define what it means for a grammar to be “ambiguous”**
- **use two additional rules for transforming a grammar**

Parse trees

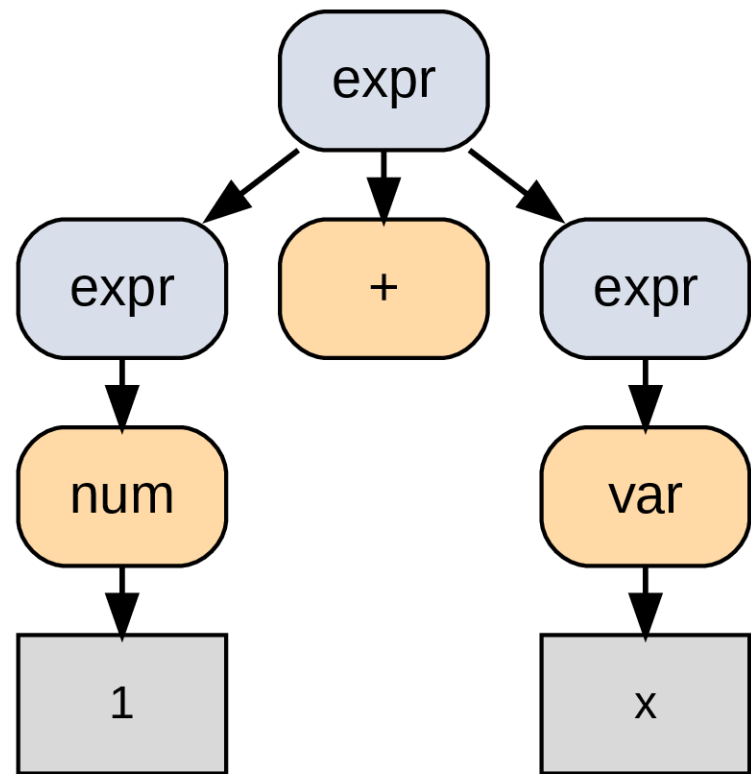
`expr ::= num | var | expr+expr`

Parse 1 + x

Parse trees

expr ::= num | var | **expr+expr**

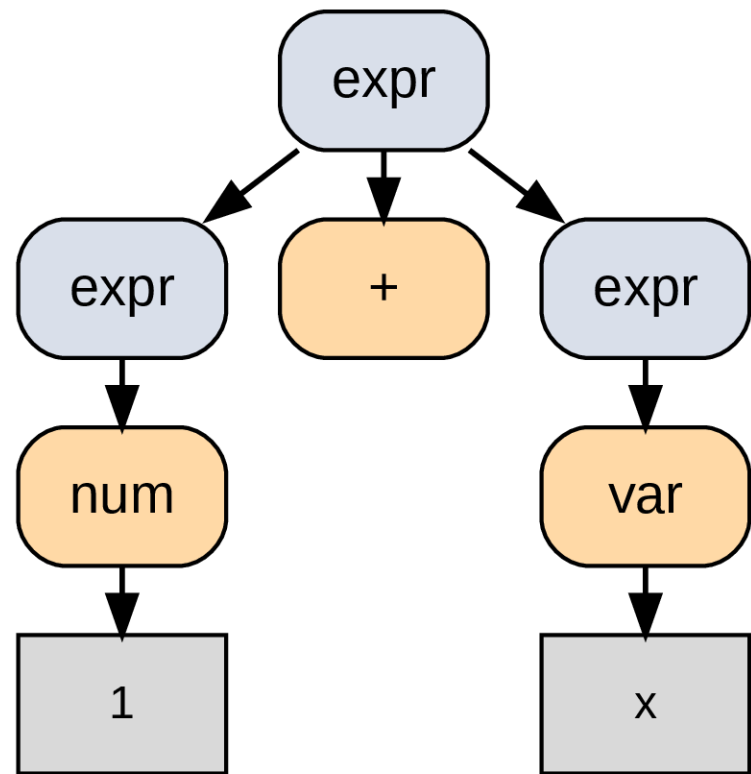
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Parse trees

expr ::= num | var | **expr+expr**

Parse 1 + x



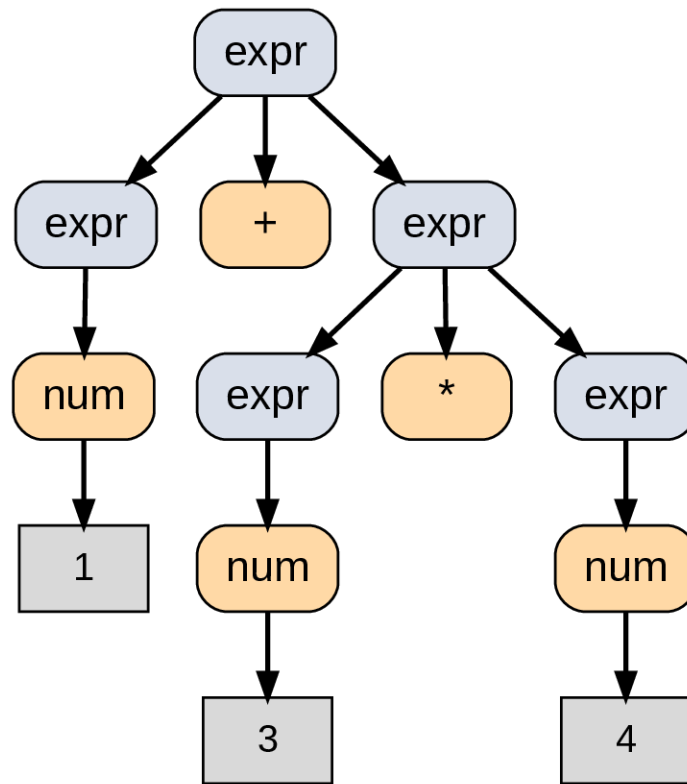
Rules:

- 1. root is labeled with the start symbol**
- 2. the symbols in one of its productions become child nodes**
- 3. continue until all leaf nodes are terminal symbols**

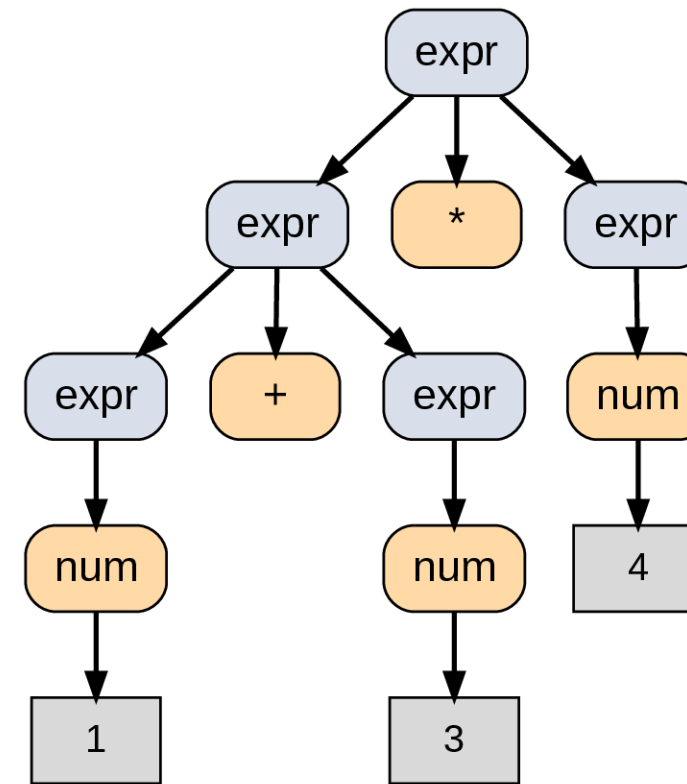
Ambiguous grammar

$\text{expr} ::= \text{num} \mid \text{expr} + \text{expr} \mid \text{expr} * \text{expr}$

Create a parse tree for $1 + 3 * 4$:



$1 + (3 * 4)$



$(1 + 3) * 4$

Ambiguous grammar

- **A grammar is ambiguous if, for some string in its language, there is more than one parse tree.**
- **Different parse trees usually suggest different meanings.**
- **So: we usually want unambiguous grammars.**

Review: predictive parsing

```
stmt ::= print(expr) | id=expr  
expr ::= id | num
```

- **We need to be able to pick a production of a non-terminal based on the next token.**

```
1 void stmt() {  
2     switch (token) {  
3         case PRINT:  
4             match(PRINT); match("("); expr(); match(")");  
5             break;  
6         case ID:  
7             match(ID); match("="); expr();  
8             break;  
9         default:  
10            error("syntax error");  
11     }  
12 }
```

Transforming a BNF grammar

- In the last lecture we learned how to deal with:
- **left recursion**
- **empty productions**

Two more ways to transform BNF:

- **left factoring**
- **expanding a non-terminal**

It's important to get comfortable with modifying grammars.

Left factoring

$A ::= aB \mid aC \mid bD$

Left factoring

$A ::= aB \mid aC \mid bD$

How about this replacement? Is it the same?

$A ::= aE \mid bD$
 $E ::= B \mid C$

Exercise

Suppose an expression can be a variable or a function call.

- **Examples:** `tot` (variable), `max(y)` (function call)

BNF:

```
expr ::= var(expr) | var
```

Exercise: Apply left factoring.

Exercise

Suppose an expression can be a variable or a function call.

- **Examples:** `tot` (variable), `max(y)` (function call)

BNF:

```
expr ::= var(expr) | var
```

Exercise: Apply left factoring.

Solution

```
expr ::= var expr1  
expr1 ::= (expr) | ""
```


Expanding a non-terminal

```
A ::= aB | E  
E ::= cC | dD
```

Any problems with a predictive parser here?

Expanding a non-terminal

```
A ::= aB | E  
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```

Any problems with a predictive parser here?

Solution

```
A ::= aB | cC | dD
```


Rewriting BNFs

- For predictive parsers, we want that, for every non-terminal in the grammar:
- there's only one production for it, or
- each production starts with a terminal token (maybe `"""` on its own) and each of those tokens are different

Exercise

```
prog ::= stmt | stmt; prog  
stmt ::= ID=expr | ID(expr)
```

- There are two problems. What are they?
- Apply left factoring.

Solution

```
prog ::= stmtprog1
prog1 ::= ;prog | ""
stmt ::= IDstmt1
stmt1 ::= =expr | (expr)
```

Exercise

Expand a non-terminal (inline `stmt` into `prog`):

```
prog ::= stmtprog1
prog1 ::= ;prog | ""
stmt ::= IDstmt1
stmt1 ::= =expr | (expr)
```

Solution

```
prog ::= IDstmt1prog1  
prog1 ::= ;prog | ""  
stmt1 ::= =expr | (expr)
```

Note: there is no longer a `stmt` non-terminal.

Fixed grammar

When expanding non-terminals, verify you didn't change the language.

Example check:

- **Original** `prog ::= stmt | stmt ; prog` **does not derive** `ID = expr ;` **(no empty statement at the end).**
- **So the transformed grammar should not allow a trailing ; either.**

```
prog ::= IDstmt1prog1
prog1 ::= ;prog | ""
stmt1 ::= =expr | (expr)
```



```

Derive ID = expr ;

grammar 1:
prog ::= stmt | stmt ; prog
stmt ::= ID = expr | ID ( expr )

prog -> stmt -> ID = expr -> FALSE
prog -> stmt ; prog -> ID = expr ; prog -> FALSE
Can't produce a ; at the end. No empty statement.

|
If we change      stmt ; prog      to      stmt ; prog1
Then we make prog1 ::= "" | stmt
I believe they will be the same then. Deriving the solution below.

NEW SOLUTION WITH THE ADDITION OF RULES
prog -> stmt ; prog1 -> ID = expr ; prog1 -> ID = expr ; = TRUE

grammar 2:
prog ::= ID stmt1 ; prog1
prog1 ::= "" | ID stmt
stmt ::= = = expr | ( expr )

prog -> ID stmt1 ; prog1 -> ID = expr ; prog1 -> ID = expr ; -> TRUE

If we get rid of "" in prog1 for grammar 2 then it would be the same.
I have derived the new test below

NEW SOLUTION WITH REDUCTION OF RULES
prog -> ID stmt1 ; prog1 -> ID = expr ; prog1 -> FALSE

```

Fixed Grammar by an old TA

Summary

- **A parse tree shows how a string can be derived from a BNF grammar.**
- **To use predictive parsing, our BNF syntax must be in a certain form. Tools to get BNF in that form:**
 - **eliminate left recursion**
 - **left factoring**
 - **expanding non-terminals**